

Document Layout Segmentation

MANGO AI

Semantic Segmentation

One of the challenges in Machine Vision is **Semantic Segmentation**, i.e., to segment an image into meaningful regions, such as Trees, Cars, or People.

Particularly, Semantic Segmentation is also highly important in Document Layout Analysis as to indentify the important regions of a Document Image, such as Tables, Titles or Lists.

Semantic Segmentation involves two specific tasks: **Spatial Identification** and **Categorical Identification**.

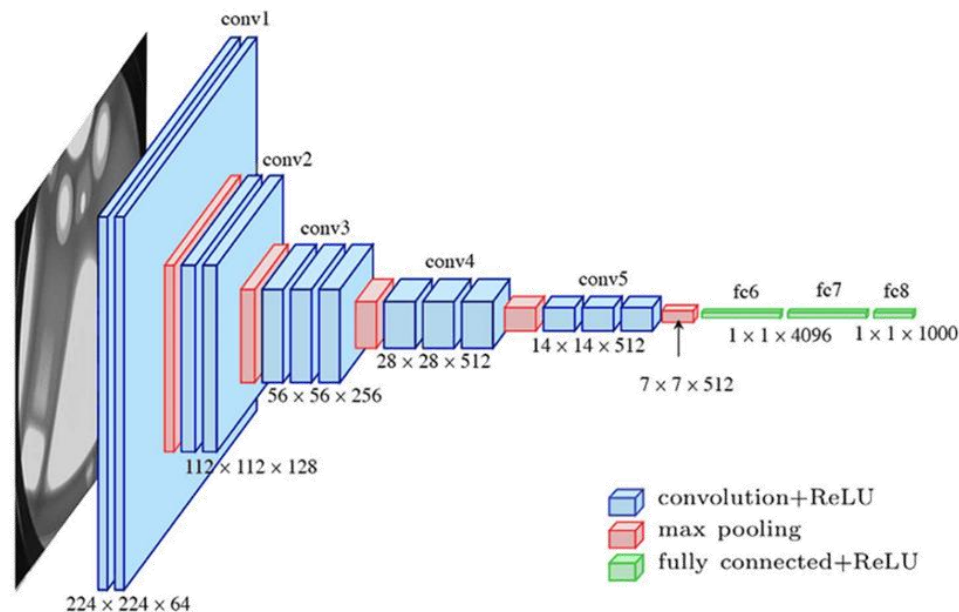
In other words, Semantic Segmentation is to identify **what is where** in the document image.

To solve this specific problem, we have come up with a Document Semantic Segmentation Model.

Segmentation Model

Our base-line model is **VGG V-NET** Model, which has been trained on **IBM DocLayout Dataset**¹.

However, since we would have to train for semantically bilingual regions, we have also employed our own Segmentation Dataset, created from our Annotation Model.



Annotation Model



Testing Company (1234567890)
 No 156, Bo Aung Gyaaw Street,
 Kyaokada A/C, Yangon, Myanmar.
 Phone: 95-1-243356 Fax: 95-1-253270 email: testing@gmail.com

Quotation

QUOTATION NO : QT-00015	DATE : 06/12/2016
NAME : ALPHA & BETA COMPUTER	REF NO :
ATTN : MR ALPHA	SALE REP : MR ALPHA
PHONE & EMAIL : +95 1 251 299 ; alphabeta@gmail.com	CURRENCY : usdusd
PARTICULAR : 2 Year renew	VALIDITY : 45 Days

No	Description	Qty	Price/Unit	Sub Total (USD)
1	iphone 7	2.00	1,234,567.00	2,469,134.00
2	iphone 6	1.00	45,678.00	45,678.00

Thank You for Your Business.
TERMS and CONDITIONS :
Payment Terms :
100% of Total to be paid upon Delivery

Total	2,514,812.00
Tax 5%	150,888.72
Total Amount:	2,665,700.72

NOTES:
(1) Price does not include Sales or Value Added Tax
 Testing Company , reserves the right to make technical changes and to correct errors in the quotation.
 The above proposal is only preliminary and subject to confirmation by signing below by both Client and Testing Company.
Delivery & Installation :
Lead Time for Payment : (1) Month **Not Available**
Installation Time

Acceptance :
 We hereby accept this offer subject to the above terms and conditions and authorize Testing Company , to treat this as our official purchase order.
 Accepted for and on behalf of :

Offer :
 For and Behalf Testing Company
 Accepted :

Name of Company
Signature & Company Stamp

Name / Title
Date



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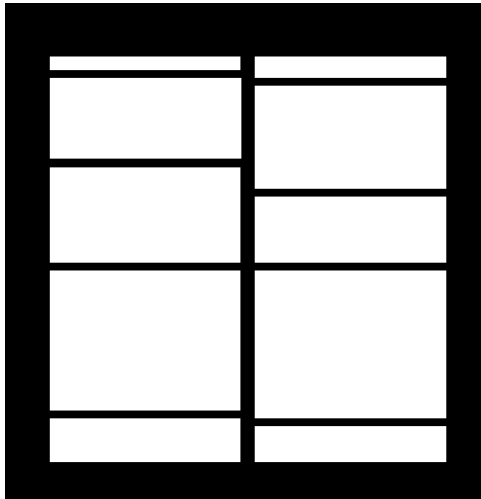
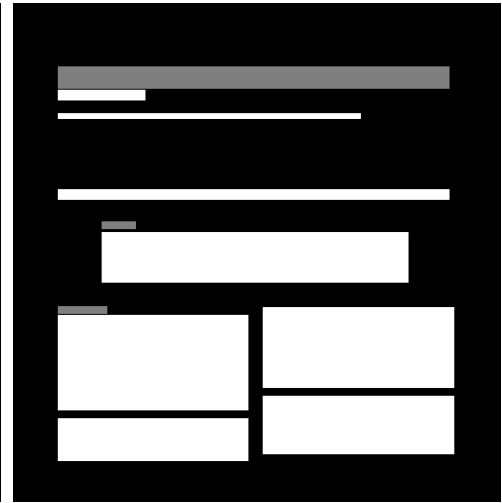
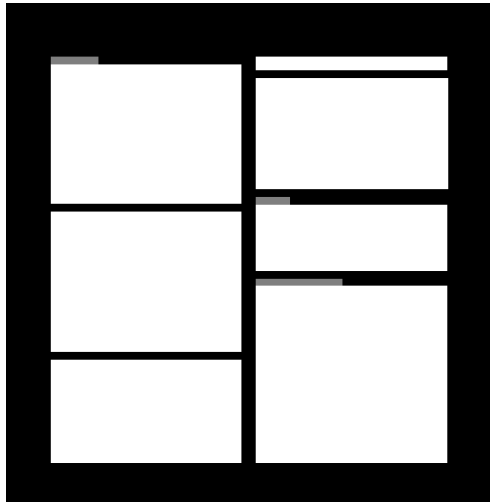
Analyze

Assign the Label to each part of the Image Segments.

☐ L-0
 ☐ L-1
 ☒ L-2
 ☐ L-3
 ☐ L-4
 ☐ L-5
 ☐ L-6
 ☐ L-7
 ☐ L-8
 ☐ L-9

0

Sample Dataset



Loss Function

Since the labels are encoded with one-hot representations, it should be possible to use simple **Cross Entropy Categorical Loss**.

However, it turns out that if we simply use **Cross Entropy Categorical Loss**, the gradient will not converge properly.

As the dataset primarily involves with a lot of “Blank” labels, i.e., Zero, there are **Class Imbalances** among dataset.

One of the earliest attempts to solve **Class Imbalance** problem is known OHEM¹ (Online Hard Example Mining), which is to sample hard examples to compute the loss after each batch.

However, we would employ **Categorical Focal Loss**, which seems more appropriate for our case.

Categorical Focal Loss

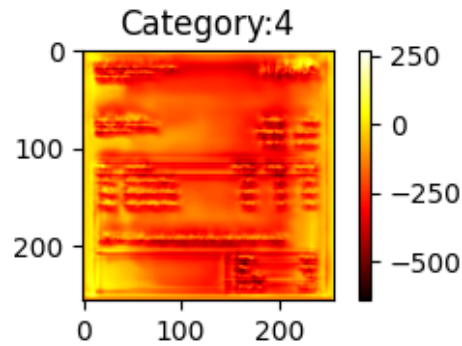
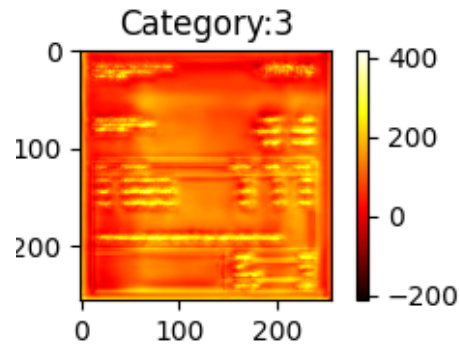
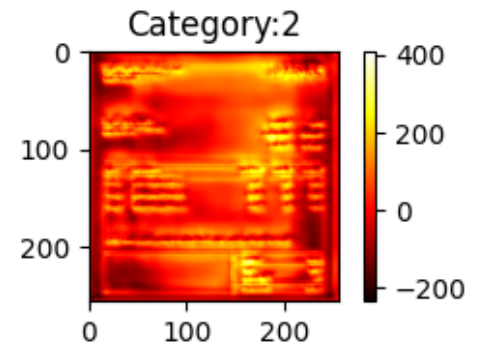
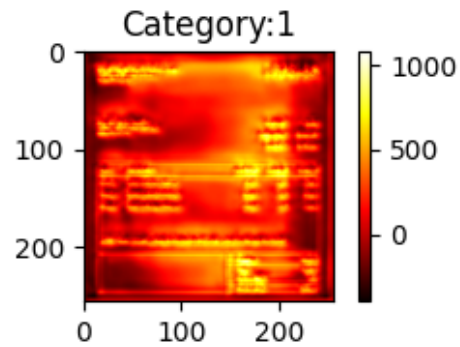
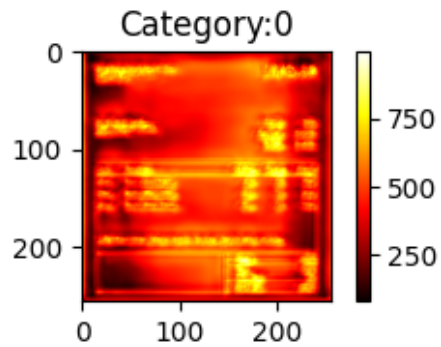
Categorical Focal Loss¹ can actually penalize the easy samples, which are zeros, and pay more attention to hard samples.

$$Loss = - \sum_{i=1}^n \alpha_i (1 - p_i)^{\gamma} \log_b(p_i)$$

where α is Weight Factor, and γ is Modulating Factor.

1. https://www.tensorflow.org/api_docs/python/tf/keras/losses/CategoricalFocalCrossentropy

Categorical Heatmap



Non-Maximum Suppression

However, from Heatmap to generating Bounding Boxes involves defining possible semantic regions.

As there are many possible regions to segment, we employ a Suppression Algorithm similar to **Non-Maximum Suppression**.

The image displays three versions of an invoice form, each with different colored bounding boxes highlighting various semantic regions. The invoice details are as follows:

Header: Example Business Name or Business Center, 123 Example Business Address, Boston, MA 02135

Invoice #: 123456

Invoice Date: 08/15/2020

Due Date: 09/15/2020

Item	Description	Unit Price	Quantity	Amount
Service	Example of service in industry	25.00	4.00	100.00
Product	Example of product in industry	800.00	1.00	800.00
Discount	Example of discount in industry	-100.00	1.00	-100.00

Footnote: (b)(7)(F), Provide a concise, professional description of the services, product, and discount listed above.

Subtotal	508.00
Total Amount Paid	508.00
Balance Due	508.00

The three versions show different segmentations: the first uses a mix of colors (green, blue, pink, purple, brown, red, teal); the second uses a mix of colors (red, purple, blue, green, yellow); and the third uses a mix of colors (green, blue, red, black, red).

CASE-1

Example Business Name or Business Owner
123 Example Business Address
Boston, MA 02135

INVOICE

Example Customer Name
456 Example Customer Address
Boston, MA 02135

Invoice # 123456
Invoice Date 08/19/2020
Due Date 09/19/2020

Item	Description	Unit Price	Quantity	Amount
Service	Example of service in industry	25.00	4.00	100.00
Product	Example of product in industry	500.00	1.00	500.00
Discount	Example of discount in industry	-100.00	1.00	-100.00
<u>NOTES:</u> Provide a concise, professional description of the services, product, and discount listed above.				
		Subtotal	600.00	
		Total	500.00	
		Amount Paid	0.00	
		Balance Due	\$500.00	

Bounding Box

Example Business Name or Business Owner
123 Example Business Address
Boston, MA 02135

INVOICE

Example Customer Name
456 Example Customer Address
Boston, MA 02135

Invoice # 123456
Invoice Date 08/19/2020
Due Date 09/19/2020

Item	Description	Unit Price	Quantity	Amount
Service	Example of service in industry	25.00	4.00	100.00
Product	Example of product in industry	500.00	1.00	500.00
Discount	Example of discount in industry	-100.00	1.00	-100.00

NOTES: Provide a concise, professional description of the services, product, and discount listed above.

		Subtotal	600.00	
		Total	500.00	
		Amount Paid	0.00	
		Balance Due	\$500.00	

CASE-2



Testing Company (123456-x)
No (56), Bo Aung Gya Street,
Kyuaktada A/C, Yangon, Myanmar.
Phone: 95-1-243556 Fax: 95-1-253270 email: testing@gmail.com

Quotation

QUOTATION NO :	QT-00015	DATE :	06/12/2016
NAME :	ALPHA & BETA COMPUTER	REF NO :	
ATTN :	MR ALPHA	SALE REP :	MR ALPHA
PHONE & EMAIL :	+95 1 251 299 ; alphabeta@gmail.com	CURRENCY :	----
PARTICULAR :	2 Year renew	VALIDITY :	45 Days

No	Description	Qty	Price/Unit	Sub Total (MYAT)
1	iphone 7	2.00	1,234,567.00	2,469,134.00
2	iphone 6	1.00	45,678.00	45,678.00

Thank You for Your Business.

TERMS and CONDITIONS :

Payment Terms :
100% of Total to be paid upon Delivery

Validity :
(7) days from Quotation Date or While Stock Last

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Delivery & Installation :
Lead Time for Equipment : (1) Week
Installation Time : Not Available

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Accepted for and on behalf of :

Name of Company
Signature & Company Stamp

Name / Title
Date

Offer :

For and Behalf Testing Company

Accepted :

Name of Company
Signature & Company Stamp

Name / Title
Date

Total 2,514,812.00
Tax 5% 150,888.72
Total Amount **2,665,700.72**

Bounding Box

Testing Company (123456-x)
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Name of Company
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Name / Title
Date

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